

probabilistic inference

CSCI
270

let's make some variables!



Xena

X

Xena's bloodtype

$X_{\text{♀}}$

gene that Xena inherited from her mother

$X_{\text{♂}}$

gene that Xena inherited from her father



Yves

Y

Yves's bloodtype

$Y_{\text{♀}}$

gene that Yves inherited from his mother

$Y_{\text{♂}}$

gene that Yves inherited from his father



Zelda

Z

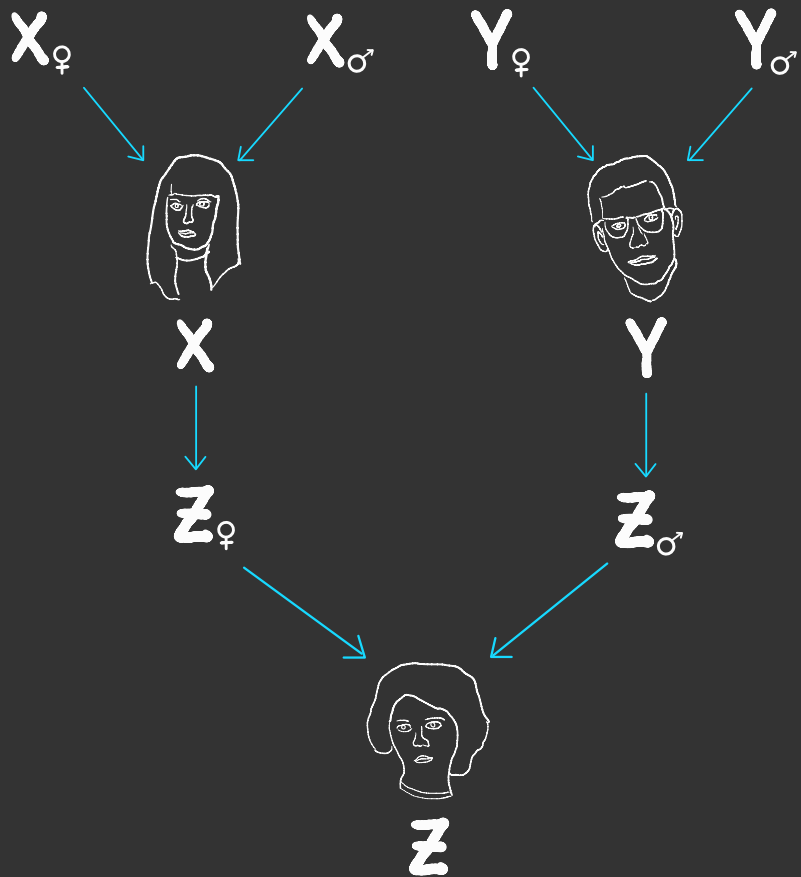
Zelda's bloodtype

$Z_{\text{♀}}$

gene that Zelda inherited from her mother

$Z_{\text{♂}}$

gene that Zelda inherited from her father

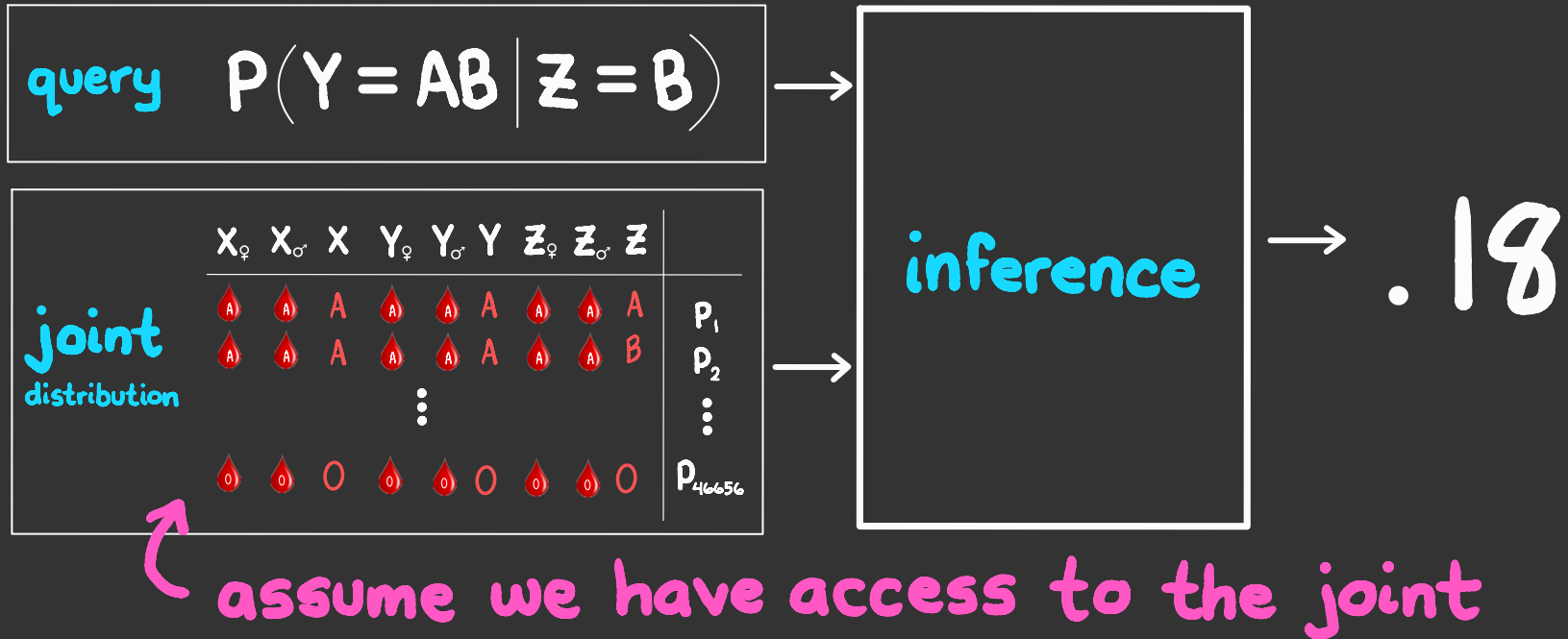


what is the probability
that Yves is type AB,
given we learn that
Zelda is type B?

in other words:

$$P(Y = AB \mid Z = B) = ?$$

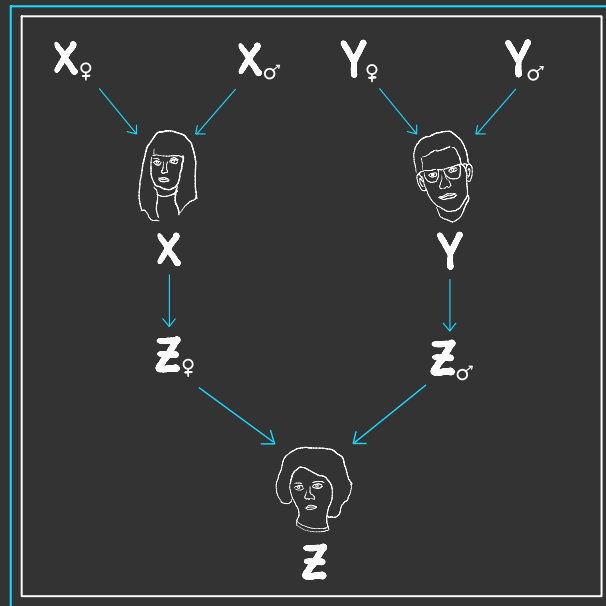
the task of answering such questions
is called **probabilistic inference**



what is the probability that Yves is type AB,
given we learn that Zelda is type B?

$$P(Y = AB \mid Z = B)$$

$$\boxed{y|x} \frac{P(Y = AB, Z = B)}{P(Z = B)}$$



$$\text{total} = \frac{\sum_{x_f} \sum_{x_m} \sum_x \sum_{y_f} \sum_{y_m} \sum_{z_f} \sum_{z_m} P(x_f, x_m, x, y_f, y_m, Y = AB, z_f, z_m, Z = B)}{\sum_{x_f} \sum_{x_m} \sum_x \sum_{y_f} \sum_{y_m} \sum_y \sum_{z_f} \sum_{z_m} P(x_f, x_m, x, y_f, y_m, y, z_f, z_m, Z = B)}$$

what is a **drawback** of this approach?

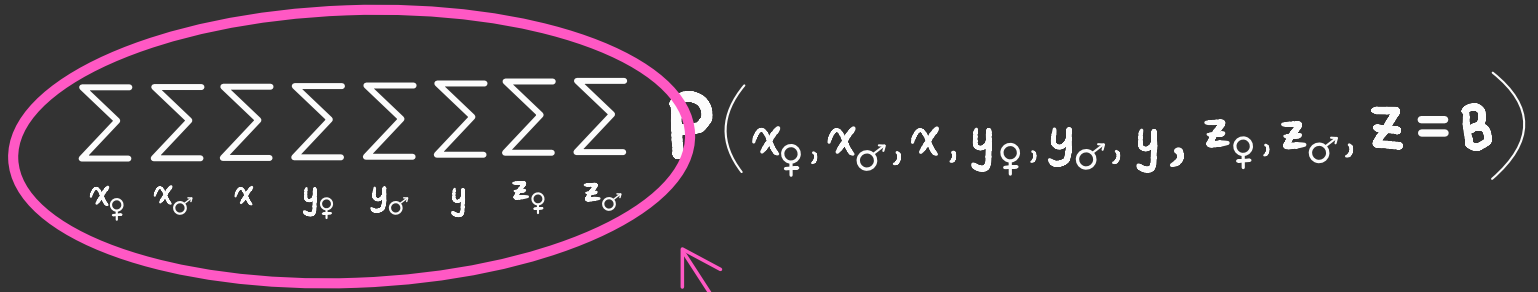
$$P(Y=AB \mid Z=B) = \frac{\sum_{x_\varphi} \sum_{x_\sigma} \sum_x \sum_{y_\varphi} \sum_{y_\sigma} \sum_{z_\varphi} \sum_{z_\sigma} P(x_\varphi, x_\sigma, x, y_\varphi, y_\sigma, Y=AB, z_\varphi, z_\sigma, Z=B)}{\sum_{x_\varphi} \sum_{x_\sigma} \sum_x \sum_{y_\varphi} \sum_{y_\sigma} \sum_y \sum_{z_\varphi} \sum_{z_\sigma} P(x_\varphi, x_\sigma, x, y_\varphi, y_\sigma, y, z_\varphi, z_\sigma, Z=B)}$$

what is a **drawback** of this approach?

$$P(Y=AB \mid Z=B) = \frac{\sum_{x_\varphi} \sum_{x_\sigma} \sum_x \sum_{y_\varphi} \sum_{y_\sigma} \sum_{z_\varphi} \sum_{z_\sigma} P(x_\varphi, x_\sigma, x, y_\varphi, y_\sigma, Y=AB, z_\varphi, z_\sigma, Z=B)}{\sum_{x_\varphi} \sum_{x_\sigma} \sum_x \sum_{y_\varphi} \sum_{y_\sigma} \sum_y \sum_{z_\varphi} \sum_{z_\sigma} P(x_\varphi, x_\sigma, x, y_\varphi, y_\sigma, y, z_\varphi, z_\sigma, Z=B)}$$

so many summations

what is a **drawback** of this approach?

$$\sum_{x_f} \sum_{x_o} \sum_x \sum_{y_f} \sum_{y_o} \sum_y \sum_{z_f} \sum_{z_o} \mathbf{P}(x_f, x_o, x, y_f, y_o, y, z_f, z_o, \mathbf{z} = \mathbf{B})$$


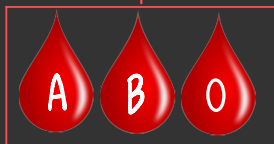
so many summations

4 values



$$\sum_{x_{\text{f}}} \sum_{x_{\text{m}}} \sum_{x} \sum_{y_{\text{f}}} \sum_{y_{\text{m}}} \sum_{y} \sum_{z_{\text{f}}} \sum_{z_{\text{m}}}$$

$$P(x_{\text{f}}, x_{\text{m}}, x, y_{\text{f}}, y_{\text{m}}, y, z_{\text{f}}, z_{\text{m}}, \mathbf{z} = \mathbf{B})$$

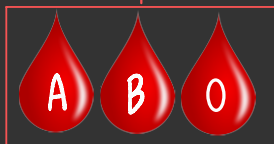


3 values

4 values



$$\begin{matrix} 3 & 3 & 4 & 3 & 3 & 4 & 3 & 3 \\ \Sigma & \Sigma & \Sigma & \Sigma & \Sigma & \Sigma & \Sigma & \Sigma \\ x_{\text{f}} & x_{\text{m}} & x & y_{\text{f}} & y_{\text{m}} & y & z_{\text{f}} & z_{\text{m}} \end{matrix}$$



3 values

$$P(x_{\text{f}}, x_{\text{m}}, x, y_{\text{f}}, y_{\text{m}}, y, z_{\text{f}}, z_{\text{m}}, \mathbf{z} = \mathbf{B})$$

11664 terms

$$\sum_{x_{\text{f}}} \sum_{x_{\text{o}}} \sum_{x} \sum_{y_{\text{f}}} \sum_{y_{\text{o}}} \sum_{y} \sum_{z_{\text{f}}} \sum_{z_{\text{o}}} \overset{3 \times 3 \times 4 \times 3 \times 3 \times 4 \times 3 \times 3}{\phantom{\sum_{x_{\text{f}}} \sum_{x_{\text{o}}} \sum_{x} \sum_{y_{\text{f}}} \sum_{y_{\text{o}}} \sum_{y} \sum_{z_{\text{f}}} \sum_{z_{\text{o}}}}$$

$$P(x_{\text{f}}, x_{\text{o}}, x, y_{\text{f}}, y_{\text{o}}, y, z_{\text{f}}, z_{\text{o}}, \mathbf{z} = \mathbf{b})$$

also the joint distribution can be
very large

query $P(Y=AB \mid Z=B)$



inference



.18

joint
distribution

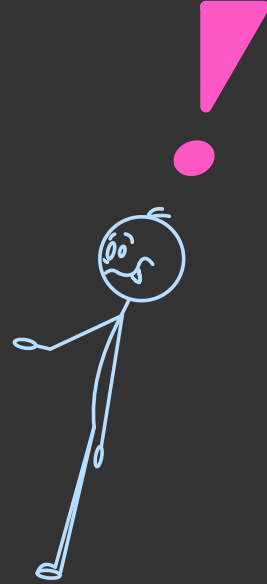
X_f	X_σ	X	Y_f	Y_σ	Y	Z_f	Z_σ	Z	
A	A	A	A	A	A	A	A	A	p_1
A	A	A	A	A	A	A	A	B	p_2
									$\vdots$
0	0	0	0	0	0	0	0	0	p_{46656}



46656 rows



P_1
 P_2
 $\vdots$
 P_{46656}



what can be done?